

Tayama Kirati

+977 9808501847 - tayamakirati@gmail.com - [linkedin.com/in/tayama-kirati-0359372b8/](https://www.linkedin.com/in/tayama-kirati-0359372b8/) - github.com/Tayama-Kirati Portfolio

EDUCATION

Tribhuwan University - Thapathali Campus

Bachelor of Engineering in Computer Engineering

2024 - 2026

Tribhuwan University - Purwanchal Campus

Bachelor of Engineering in Computer Engineering

2022 - 2024

St. Xaviers College, Maitighar

+2 Science

2019 - 2021

TECHNICAL SKILLS

Programming Languages: C/C++, Python, JavaScript, TypeScript, SQL

Frameworks & Libraries: React, Next.js, Express, NodeJS, NumPy, Pandas, PyTorch, scikit-learn

Databases: SQLite, PostgreSQL, MongoDB

Tools & Platforms: Hadoop, Figma, Unity, Blender

PROJECTS

Text-to-ASL Conversion System (Minor Project)

github.com/bhattaraisushma/Snapsigns

3D Sign Language Translation Framework

Python, NLP, T5, Blender

- Designed and developed a text-to-ASL translation system to convert written text into American Sign Language animations.
- Utilized the T5 transformer model for natural language understanding and sentence to sign mapping.
- Created 3D animated ASL gestures using Blender for accurate and expressive sign visualization.
- Built an end-to-end pipeline integrating NLP processing with real-time 3D animation rendering.

Monocular SLAM and NeRF Fusion for 3D Reconstruction (Major Project)

Real-Time Neural Rendering and Localization

Python, OpenCV, SLAM, NeRF

- Built a geometric SLAM pipeline generating optimized poses and sparse point clouds from monocular video.
- Integrated Depth Anything V2 to produce detailed depth maps, for accurate 3D reconstruction.
- Established a data pipeline that bridges SLAM and NeRF by providing optimized poses and dense depth maps for neural rendering.
- Training a real-time NeRF to synthesize photorealistic, dynamic 3D scenes from the SLAM and depth outputs.

MetMoMo

github.com/Tayama-Kirati/MetMoMo

Full-Stack Food Ordering Web Application

React, Node.js, Express, MongoDB, Cloudinary

- Built a full-stack food ordering platform with a decoupled frontend and backend architecture.
- Implemented JWT-based authentication and secure user sessions for customers and restaurant owners.
- Integrated Cloudinary for product and restaurant image uploads via Multer, storing URLs in MongoDB.
- Designed RESTful APIs with Express and Node.js to handle orders, menus, and user management.

E-Commerce

github.com/Tayama-Kirati/E-Commerce

Modern E-Commerce Web Application

Next.js, TypeScript, Prisma

- Developed a full-featured e-commerce website using Next.js with TypeScript for type-safe development.
- Used Prisma ORM for database schema management and efficient data querying.
- Leveraged Next.js App Router for server-side rendering and optimized page load performance.

My Space

github.com/Tayama-Kirati/My-Space

Personal Digital Content Management App

JavaScript, Node.js, CSS

- Built a personal content management site featuring a daily journal, book tracker, photo memory gallery, and focus sessions.
- Designed an interactive journal with date-based entry editing, monthly cover photos, and a year overview with 12 mini month cards.
- Implemented a book logging system with star ratings, personal reviews, and a stats dashboard showing reading progress.
- Created a masonry photo gallery with drag-and-drop upload, captions, mood tags, and a monthly timeline sidebar.

Rock Paper Scissors AI

github.com/Tayama-Kirati/RockPaperScissors

Adaptive AI Game using Markov Chains

Python

- Built a terminal-based game where an AI learns and predicts the player's move patterns using a Markov Chain transition table.
- Implemented persistent AI memory using JSON so the model retains player history across sessions.
- Designed modular OOP architecture with separate classes for AI brain, game logic, and statistics tracking.

EXPERIENCE

Fellowship, All In Foundation (AIF)

Feb 2025 – Sep 2025

- Produced a documentary on the Dom community, highlighting their cultural identity, social challenges, and lived experiences. Conducted field research and interviews to document underrepresented voices, ensuring authentic and respectful representation through storytelling

Internship, E-Digital Nepal

May 2026 - Present

- Designed UI/UX for Digital Nepal ERP – Reimagine, including wireframes, prototypes, and a reusable Figma component library
- Developed responsive frontend features (React, Next.js, Tailwind CSS, TanStack) for modules including Dynamic Form Builder, Academics, Student Employee Management, and Authentication
- Collaborated with cross-functional teams to translate approved designs into pixel-perfect, functional interfaces